MA104 · DISCRETE MATHEMATICS

Assignment 03 — Mathematical Induction

Questions: **18**
Due: ___ / ___ / **2026**

Chapter 3 — Lecture 3 with the exercise sheets E02 and E03

Name: ..... **ID:** .....

Group: ..... **Date:** .....

Instructions. Work through the parts in order; each part builds on the one before it.

- Every induction proof must show **three labelled parts**: the basis step (computed, not assumed), the inductive hypothesis (for one k), and the inductive step.
- In the inductive step, **mark the line where you use the hypothesis** — write "(by the IH)".
- Start the inductive step from **one** side and transform it into the other. Do not manipulate both sides of the equation you are trying to prove.
- Hand-write your answers on this sheet, in the space given.
- **Answer without AI.** ChatGPT, Claude, Gemini and similar tools are not allowed, and neither is copying from a classmate or from the solved sheets. You may be asked to explain any answer at the board, and an answer you cannot explain will not be accepted.

The principle, in one line

$$P(1) \quad \wedge \quad \forall k \geq 1 \; (P(k) \to P(k+1)) \quad \Longrightarrow \quad \forall n \geq 1 \; P(n)$$

If the statement starts at $n = n_0$ instead of 1, the basis step checks $P(n_0)$ and the inductive step is proved for every $k \geq n_0$.

Part A — Complete the Proof

Lecture 3, Examples 1, 5 and 7. Fill in each blank.

Question 1 cf. L03 Ex. 1

Prove that the sum of the first n odd numbers is n^2:

$$1 + 3 + 5 + \cdots + (2n-1) = n^2, \quad n \geq 1$$

Basis step. For $n = 1$: LHS = _____, RHS = _____, so the formula holds for $n = 1$.

Inductive hypothesis. Assume that for some $k \geq 1$, $1 + 3 + \cdots + (2k-1) =$ _____.

Inductive step. The next odd number is $2(k+1) - 1 =$ _____. Adding it to both sides of the hypothesis,

$$1 + 3 + \cdots + (2k-1) + (2k+1) = _____ + _____ = _____,$$

which is the formula with $n = k + 1$. By induction it holds for every $n \geq 1$.

Question 2 cf. L03 Ex. 5

Prove that the sum of the first n even numbers is $n(n+1)$:

$$2 + 4 + 6 + \cdots + 2n = n(n+1), \quad n \geq 1$$

Basis step. For $n = 1$: LHS = _____, RHS = _____.

Inductive hypothesis. Assume $2 + 4 + \cdots + 2k =$ _____.

Inductive step. Adding the next term _____ to both sides,

$$\begin{aligned} 2 + 4 + \cdots + 2k + 2(k+1) &= _____ + 2(k+1) \\ &= (k+1)(_____) \quad \text{factor out } (k+1), \end{aligned}$$

which is $n(n+1)$ with $n = k+1$.

Question 3 E02 Q2 · cf. L03 Ex. 7

Prove that $6^n - 1$ is divisible by 5 for every positive integer n.

Basis step. For $n = 1$: $6^1 - 1 =$ _____, which is divisible by 5.

Inductive hypothesis. Assume _____ is divisible by 5, i.e. $6^k - 1 = 5m$ for some integer m.

Inductive step. Split $6 = 5 + 1$:

$$6^{k+1} - 1 = 6 \cdot 6^k - 1 = _____ \cdot 6^k + (_____)$$

The first term is divisible by 5 because _____; the second is divisible by 5 by _____. Hence $6^{k+1} - 1 = 5\big(6^k + m\big)$ is divisible by 5.

Part B — Summation Formulas

Lecture 3, Examples 1–5; E02 Q1, Q3; E03 Ex. 2–5. Write a complete proof for each.

Question 4 E02 Q1 · E03 Ex. 5

Prove that for all $n \geq 1$,

$$1 + 4 + 7 + \cdots + (3n - 2) = \frac{n(3n-1)}{2}$$

Basis step.

Inductive hypothesis.

Inductive step.

Question 5 E03 Ex. 2 · cf. L03 Ex. 5

Prove that for all $n \geq 1$,

$$3 + 3^2 + 3^3 + \cdots + 3^n = \frac{3^{n+1}-3}{2}$$

Basis step.

Inductive hypothesis.

Inductive step.

Question 6 E03 Ex. 3

Prove that for all $n \geq 1$,

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

Basis step.

Inductive hypothesis.

Inductive step.

Question 7 E03 Ex. 4

The triangular numbers are $1, 3, 6, 10, \ldots$ with n-th term $\frac{n(n+1)}{2}$. Prove that for all $n \geq 1$,

$$1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

Basis step.

Inductive hypothesis.

Inductive step.

Question 8 E02 Q3

Prove that the sum of the squares of the first $2n$ positive integers is

$$1^2 + 2^2 + 3^2 + \cdots + (2n)^2 = \frac{n(2n+1)(4n+1)}{3}, \quad n \geq 1$$

Basis step.

Inductive hypothesis.

Inductive step.

Question 9 cf. L03 Ex. 3, 4

Prove that for all $n \geq 1$,

$$\frac{1}{1\cdot 4} + \frac{1}{4\cdot 7} + \frac{1}{7\cdot 10} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

Basis step.

Inductive hypothesis.

Inductive step.

Part C — Divisibility

Lecture 3, Examples 6–7; E02 Q2. Rewrite $f(k+1)$ so that $f(k)$ appears inside it.

Question 10 cf. L03 Ex. 7

Prove that $4^n - 1$ is divisible by 3 for every $n \geq 1$.

Basis step.

Inductive hypothesis.

Inductive step.

Question 11 cf. L03 Ex. 7

Prove that $3^{2n} - 1$ is divisible by 8 for every $n \geq 1$.

Basis step.

Inductive hypothesis.

Inductive step.

Question 12 cf. L03 Ex. 6

Prove that $n^3 + 5n$ is divisible by 6 for every $n \geq 1$.

Basis step.

Inductive hypothesis.

Inductive step.

Part D — Inequalities and Recursive Sequences

E02 Q4–Q7. Watch the starting value of n, and how many earlier cases each step needs.

Question 13 E02 Q5

Prove that $n! > 3^n$ for every $n \geq 7$.

Basis step.

Inductive hypothesis.

Inductive step.

Question 14 E02 Q4

A sequence is defined by

$$x_1 = 1 \quad \text{and} \quad x_{n+1} = 1 + \sqrt{2x_n} \quad \text{for } n \geq 1.$$

Prove that $x_n < 4$ for all $n \geq 1$.

Basis step.

Inductive hypothesis.

Inductive step.

Question 15 E02 Q7

The Fibonacci sequence is defined by $x_1 = 1$, $x_2 = 1$ and $x_n = x_{n-1} + x_{n-2}$ for $n \geq 3$.

(a) Complete the table and give x_{20}.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
1	1	_____	_____	_____	_____	_____	_____	_____	_____

x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{17}	x_{18}	x_{19}	x_{20}
_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

(b) Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Use **strong** induction to prove

$$x_n = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n), \quad n \geq 1.$$

Basis step.

Inductive hypothesis.

Inductive step.

(c) Use the formula of part (b) to compute x_{20}.

Question 16 E02 Q6 · bonus

Let $p_0 = 1$, $p_1 = \cos\theta$ (for a fixed constant θ) and $p_{n+1} = 2p_1 p_n - p_{n-1}$ for $n \geq 1$. Use strong induction to prove that

$$p_n = \cos(n\theta) \quad \text{for all } n \geq 0.$$

Basis step.

Inductive hypothesis.

Inductive step.

Part E — Find the Flaw

Each "proof" below reaches a false conclusion. Find the exact step that fails.

Question 17

Claim. $1 + 2 + 3 + \cdots + n = \frac{n^2+n+2}{2}$ for all $n \geq 1$.

"Proof." Assume the formula for $n = k$. Adding $k + 1$ to both sides,

$$1 + \cdots + k + (k+1) = \frac{k^2 + k + 2}{2} + (k+1) = \frac{k^2 + 3k + 4}{2} = \frac{(k+1)^2 + (k+1) + 2}{2},$$

which is the formula for $n = k + 1$. Hence the claim is true.

(a) Is the algebra of the inductive step correct?

(b) What goes wrong? Test $n = 1$.

Question 18

Claim. In every group of $n \geq 1$ horses, all the horses have the same colour.

"Proof." For $n = 1$ the claim is clearly true. Assume it holds for every group of k horses, and take a group of $k + 1$ horses $h_1, h_2, ..., h_{k+1}$. Removing h_{k+1} leaves k horses, which all share one colour by the hypothesis; removing h_1 leaves k horses, which also share one colour. The two groups overlap, so all $k + 1$ horses have the same colour.

Which step fails, and for which value of k?

— END OF ASSIGNMENT —